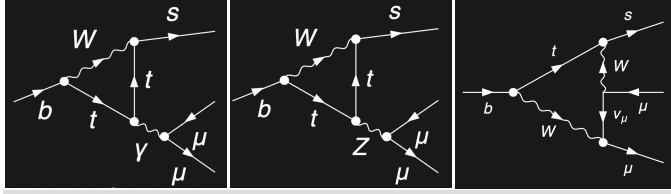


Rare Decay $B \rightarrow K^* \ell^+ \ell^-$: From Effective Field Theory to New Physics

Garvit Sonawala | SRIP (May 2026 - July 2026) IIT Gandhinagar | Guide: Dr. Rusa Mandal

Feynman diagram to H_{eff}



Penguin and box diagrams for $b \rightarrow s \ell^+ \ell^-$ (top-quark dominant contribution), shown for $\ell = \mu$ as a representative case; the diagrams are identical for $\ell = e, \tau$.

- Suppressed by three independent effects: no tree-level contribution exists (**loop-level only**), a small **CKM factor** ($V_{tb} V_{ts}^*$), and an incomplete **GIM cancellation** that survives only because the top quark is far heavier than u, c
- The full amplitude is assembled from these diagrams using the Feynman rules; only the **top-quark loop** is kept, since GIM makes it dominant

$$-i\mathcal{M}_\tau = \int \frac{d^4 k}{(2\pi)^4} \bar{u}(p_s) \left[\frac{ig}{\sqrt{2}} V_{ts}^* \gamma^\mu P_L \right] \frac{i(k - \not{q} + m_t)}{(k - q)^2 - m_t^2} \times (-iQ_\ell e \gamma^\mu) \frac{i(k + m_t)}{k^2 - m_t^2} \left[\frac{ig}{\sqrt{2}} V_{tb} \gamma^\mu P_L \right] u(p_b) \\ \times \left[-i \left(g_{\mu\nu} - \frac{(p_b - k)_\mu (p_b - k)_\nu}{M_W^2} \right) \frac{1}{(p_b - k)^2 - M_W^2} \right] \frac{-ig_{\alpha\sigma}}{q^2} \bar{u}(p_1) (-ieQ_\ell \gamma^\sigma) v(p_2)$$

- This is just one of three diagrams and, like the others, individually **gauge-dependent**. Combining all three (via **heavy-mass expansion**, **Passarino-Veltman reduction**, and **Inami-Lim functions**) restores **gauge invariance**:

$$\mathcal{M}_{full} \propto V_{tb} V_{ts}^* \frac{G_F}{\sqrt{2}} \frac{\alpha}{2\pi \sin^2 \theta_W} \left\{ \left[\frac{Y_0}{\sin^2 \theta_W} - 4Z_0 \right] (\bar{s} \gamma_\sigma P_L b) (\bar{\ell} \gamma^\sigma \ell) - \left[\frac{Y_0}{\sin^2 \theta_W} \right] (\bar{s} \gamma_\sigma P_L b) (\bar{\ell} \gamma^\sigma \gamma_5 \ell) \right\}$$

- These Lorentz structures also appear in the effective theory, so the result is absorbed into two **Wilson coefficients** multiplying two **local operators**, forming H_{eff}

$$\mathcal{H}_{eff} = - \frac{4G_F}{\sqrt{2}} \frac{\alpha}{4\pi} V_{tb} V_{ts}^* [C_9 \mathcal{O}_9 + C_{10} \mathcal{O}_{10}]$$

- Matching $\mathcal{M}_{full}$ onto $\mathcal{H}_{eff}$ at $\mu \sim M_W$ fixes the **Wilson coefficients** C_9, C_{10} . These are then evolved down to $\mu \sim m_b$ via the **renormalization group**, resumming large logarithms that would otherwise wreck the calculation

Simplified derivation: top-quark contribution only, illustrating the pipeline behind the full 3-quark calculation.

What is this decay?

- A B meson transforms into a K^* meson + lepton pair ($e^+ e^-$ or $\mu^+ \mu^-$)
- Quark level: $b \rightarrow s \ell^+ \ell^-$, a flavor-changing neutral current, forbidden at tree level

Why does it matter?

- A tiny rate, but a sensitive one:** the Standard Model predicts this decay happens only about once in 10 million times; if new, undiscovered physics is at play, it would show up as a noticeably bigger change here than in more common decays
- Actively being tested right now:** experiments like LHCb and Belle II have measured this decay in detail, and some results don't quite match predictions.
- Connected to a bigger mystery:** the Standard Model can't explain why particle masses and mixing patterns look the way they do; this decay helps test the leading ideas trying to solve that

Hadronic input & long-distance effects

- C_9 and C_{10} carry the short-distance physics, but b and s quarks live inside mesons, not free space. Connecting the two means evaluating **matrix elements** like:

$$\langle \bar{K}^*(k) | \bar{s} \gamma^\mu b | \bar{B}(p) \rangle \supset \epsilon^{\mu\nu\rho\sigma} \epsilon_\nu^* p_\rho k_\sigma \frac{2V(q^2)}{m_B + m_{K^*}}$$

One of seven form factors parametrizing this decay, $V(q^2)$, from the vector current

- These can't be calculated from first principles, so they're parametrized by **form factors**, functions capturing how likely a B meson becomes a K^* meson (7 for $B \rightarrow K^*$), computed via **light-cone sum rules** (low q^2) and **lattice QCD** (high q^2)
- The naive picture assumes short-distance and long-distance pieces simply multiply together, but this breaks down once **gluon exchange** with the spectator quark is included
- QCD factorization** splits this into a calculable hard piece and a non-perturbative soft piece, plus contributions from **four-quark operators** and **final-state interactions**
- Combining Wilson coefficients with form factors gives the full **decay amplitude**, the quantity behind every observable discussed next

Observables

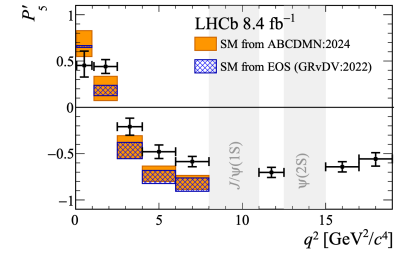
- K^* is spin-1, so its decay adds three angles on top of q^2 . These are captured by three **transversity amplitudes**: A_0 (longitudinal), $A_\parallel$, $A_\perp$ (transverse)

$$A_\perp^{L,R} = N \sqrt{2} \lambda^{1/2} \left[(C_9^{\text{eff}} + C_9^{\text{eff}}) \mp (C_{10}^{\text{eff}} + C_{10}^{\text{eff}}) \right] \frac{V(q^2)}{m_B + m_{K^*}} + \frac{2m_b}{q^2} (C_7^{\text{eff}} + C_7^{\text{eff}}) T_1(q^2)$$

One transversity amplitude ($A_\perp$), showing Wilson coefficients and form factors combined, the building block for every observable

- Most observables come from interference between these amplitudes, not any single one alone. Different observables probe different Wilson coefficients, so no single measurement can confirm new physics by itself

- P_5' is built as a ratio, so most of the form-factor uncertainty cancels out. It is also the observable where the clearest tension with the SM has shown up
- R_{K^*} compare muon vs electron decay rates. Early hints of deviation were resolved by an improved 2022 LHCb measurement, a reminder that tensions can fade with better data
- Statistics are limited, so results are reported in q^2 bins, skipping the J/ψ and $\psi(2S)$ region. Theory and experiment are then compared globally, fitting all observables at once rather than one at a time



P_5' vs q^2 , LHCb data (points) against two independent SM predictions (bands). A reproducible $\sim 3\sigma$ pull appears in the 4-8 GeV^2 region, one of the most closely watched tensions in flavor physics: genuine new physics, or underestimated hadronic effects?

Beyond the Standard Model

- The Standard Model has no explanation for why particle masses and mixing patterns look the way they do. This is the **flavor puzzle**, and it motivates looking beyond the SM
- Proposed solutions fall into three broad classes:
 - Symmetry-based** (e.g. $U(2)^3$ flavor symmetry): a symmetry restricts new physics to resemble the SM's own flavor structure
 - Geometric/dynamical** (e.g. flavor deconstruction / '4321' models): the hierarchy emerges from the structure of space or strongly-coupled dynamics itself
 - Mediator-based** (e.g. Z' bosons, leptoquarks coupling preferentially to left-handed muons): a single new particle is introduced, usually motivated by a specific anomaly rather than the hierarchy problem

Conclusion

- This project followed one rare decay, $B \rightarrow K^* \ell^+ \ell^-$, from the underlying calculation to real measurements by **LHCb** and **Belle II**
- The Standard Model predicts this decay with **extreme precision**, so any small deviation is a promising place to look for **new physics**
- One measurement, P_5' , shows a persistent $\sim 3\sigma$ gap from prediction, not yet a discovery, but one of the most closely watched tensions in the field
- Real new physics, or gaps in how we model the messier parts of the calculation? **Better data and theory will tell**